Growth at the Margin: Political Incentives and Firm Behavior in China

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Abstract

This paper examines how China's GDP growth targets shape political incentives and firm behavior. Using survival models, bunching analysis, and a threshold-based design, I show that performance relative to targets increases leaders' promotion prospects and generates bunching above the target threshold. At the firm level, meeting-target pressure produces discontinuous increases in inventories, sales, and output, especially when cities are close to their targets or have underperformed earlier. Firm-specific energy consumption and pollution exhibit similar patterns, suggesting that at least part of the response reflects real production rather than statistical manipulation alone. Growth targets affect firm-level output and resource allocation.

JEL Codes: D73, O43, D22, H11, O12

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1 Introduction

Since the onset of economic reforms in 1978, China has experienced unprecedented economic growth, with GDP expanding more than fortyfold over the past four decades. A cornerstone of this transformation has been the manufacturing sector, which has played a critical role in driving economic expansion. Scholars frequently attribute this success to China's distinctive model of economic governance, characterized by the use of growth targets to align macroeconomic performance with national priorities (Prasad and Rajan, 2006). Central to this framework is a performance-based cadre evaluation system, which closely ties local officials' career advancement to their ability to achieve economic growth within their jurisdictions, with regional GDP growth serving as a primary benchmark for promotion decisions (Chen et al., 2018; Li and Zhou, 2005). However, a growing body of research has raised concerns about the unintended consequences of the country's growth strategy—particularly its role in fueling administratively driven resource allocation, mounting local debt, and persistent imbalances in domestic demand (Hsieh and Klenow, 2009; Chen et al., 2022; Shen and Chen, 2017; Dong and Sun, 2022). One increasingly salient concern is the phenomenon of structural overcapacity and overproduction in certain sectors, where output appears to exceed demand on a sustained basis (Kenderdine and Ling, 2018; Tan and Conran, 2022; Xu and Liu, 2018).[1] While this paper does not seek to measure the macroeconomic extent of such phenomena, it contributes to understanding their institutional roots by examining a key mechanism at the local level. Specifically, it investigates how political incentives—shaped by the linkage between GDP targets and official promotions—can induce production behavior at the firm level that prioritizes administrative objectives over market conditions. By documenting systematic patterns of output increases around the threshold of target fulfillment, this paper provides micro-level evidence that political incentives can shape resource allocation, offering a possible pathway to understanding excess capacity in China's growth model.

[1] U.S. Treasury Secretary Janet Yellen, for instance, has warned that China's practices, including excessive production and exports, could pose significant threats to global market stability.

In this paper, I make three key contributions to understanding how growth-target incentives shape politicians' behavior and economic outcomes in China. First, I employ a discrete-time survival analysis using detailed data on the career trajectories of city-level officials to examine how meeting GDP growth targets affects their promotion prospects. This framework accounts for the fact that promotion probabilities vary systematically with tenure length, while also accommodating time-varying performance and right-censored career spells. Second, I construct a novel city-level panel dataset covering nearly two decades of annual growth targets for over 300 cities. Leveraging this dataset, I conduct a bunching analysis to assess how officials respond to growth-related pressures, as reflected in the distribution of performance scores—defined as the difference between actual and target GDP growth rates. The results show a strong link between target attainment and promotion outcomes: a one-unit increase in performance score raises the probability of promotion by 9–10%, and officials who meet or exceed targets tend to be promoted approximately 1.5 years earlier one average than those who fall short. I also find substantial excess mass in the distribution of performance scores around the growth threshold, with approximately 50% more observations than predicted by a smooth counterfactual distribution at the exact threshold and close to 100% excess mass when the bunching region is defined more broadly. [2] The location and shape of this excess mass admit two related interpretations. A sharp concentration at or immediately above zero may reflect reporting, rounding, or administrative discretion that places borderline cities on the successful side of the threshold. By contrast, a real behavioral response need not produce exact bunching. Officials may intensify administrative mobilization or exert greater pressure on local firms as the target comes within reach, but the mapping from such efforts to realized annual GDP growth is inherently noisy. Their responses may therefore generate diffuse excess mass over a neighborhood around zero rather than allow them to hit the threshold precisely. Accordingly, the bunching analysis reports both excess mass at the exact threshold and diffuse bunching over a broader interval around it. An al-

[2] These findings contrast with previous literature suggesting that, following China's economic reforms, growth targets serve merely as guidelines rather than binding goals (Li et al., 2019).

ternative possibility is that local governments may manipulate reported GDP to meet their targets. This concern is consistent with Martinez (2022), who documents that autocracies report systematically higher GDP growth relative to satellite-recorded night-time lights and estimates that their annual GDP growth is overstated by approximately 35%. The third empirical exercise in this paper helps distinguish this reporting channel from real economic responses.

I make a third contribution by examining how political incentives cascade from local officials to firms. Using a threshold-based empirical strategy, I investigate whether pressures to meet growth targets translate into real economic responses among local manufacturing firms—a sector central to China's economy and closely intertwined with local governments.[3] To conduct this analysis, I construct a rich panel dataset by merging the China Industrial Enterprise Database (CIED) with the China Industrial Enterprise Pollution Database (CIEPD), resulting in over 300,000 firm-year observations across more than 20 manufacturing industries. This merged dataset allows me to track both financial indicators and physical metrics such as energy consumption and pollutant emissions—two proxies far less vulnerable to manipulation. By linking firm-level outcomes with city-level target fulfillment, I uncover sharp discontinuities in output, energy use, and pollution precisely at the point where cities just meet their growth targets. These synchronized spikes indicate that a large share of the observed firm response stems from real production activity—even excessive production leading to inventory buildup—rather than mere statistical inflation. Although I cannot fully rule out the possibility of data manipulation, the use of output proxies indicates that over 80% of the observed effect reflects real economic activity. These findings reinforce the broader narrative that China's target-driven governance system generates tangible, micro-level economic effects.

Although the firm-level specification resembles a regression discontinuity design, I do not

[3]By 2020, China accounted for 20% of global manufacturing exports, cementing its status as the world's leading manufacturing powerhouse. This dominance is further reflected in the industrial manufacturing sector's contribution to the national economy, which accounted for approximately 27% to 33 of GDP in 2022.

interpret the discontinuity as the causal effect of crossing the target threshold. Such an interpretation would require the actual–target gap to be locally unmanipulated, whereas in this setting the running variable is itself shaped by the strategic responses of interest. This endogenous movement around the cutoff violates the standard RDD identifying assumption, but it is precisely the behavior the paper seeks to document. I therefore interpret the discontinuities as evidence of strategic responses to target incentives, with firm-level energy use, pollution emissions, and other physical outcomes providing corroborating evidence of changes in real economic activity.

This paper contributes to the literature on institutional target setting and government performance management, a field long debated for its benefits and unintended consequences. Targets can provide clear goals, enhance accountability, and align performance with organizational priorities (Hood, 1991; Li et al., 2019). However, they also introduce risks such as gaming—manipulating data to superficially meet benchmarks (Christopher and Hood, 2006; Zeng and Zhou, 2024)—and tunnel vision, which incentivizes narrow focus on quantifiable indicators at the expense of broader objectives (Bevan and Hood, 2006). While target-based systems promote data-driven decision-making (Smith, 1995), they may also reduce institutional adaptability (Radnor and McGuire, 2004) and demotivate actors when goals are poorly designed (Van Thiel and Leeuw, 2002).[4]

In the context of China, despite a growing body of research on the macroeconomic effects of growth targets (Zhao and Cheng, 2023; Gong et al., 2025), the mechanisms at the micro level remain insufficiently explored. Specifically, scant empirical work has investigated how promotion incentives tied to growth targets influence local officials' behavior and, in turn, affect firm-level outcomes.[5] Although several studies have raised concerns about the reliability of China's macroeconomic data (Zeng and Zhou, 2024; Firth et al., 2011; Chen et al.,

[4]Scholars emphasize that effective implementation depends on context-sensitive targets, often combined with qualitative or participatory evaluation tools (Moynihan, 2008; Pollitt and Bouckaert, 2011).

[5]Existing studies offer mixed evidence: some suggest these incentives boost innovation and administrative efficiency (Sun et al., 2023; Dai et al., 2021), while others associate them with resource misallocation and environmental harm (Liu et al., 2020).

2019),[6] suggesting that performance targets may drive strategic reporting or data manipulation just to hit or barely exceed assigned goals (Martinez, 2022), less attention has been paid to the core question addressed here—to what extent, and through which mechanisms, do growth targets translate into real economic activity? This paper demonstrates that political incentives at the local level can provoke real shifts in firm behavior, not merely superficial adjustments to official statistics.

This paper also contributes to the literature on political incentives and bureaucratic governance in hierarchical authoritarian regimes. Within China's administrative structure, the attainment of economic and political benchmarks remains a key determinant of promotion outcomes for local officials (Li and Zhou, 2005; Chen et al., 2018; Yao and Zhang, 2015). At the same time, informal institutions such as guanxi—unobservable personal networks—continue to shape bureaucratic behavior and influence both decision-making and resource allocation (Gold et al., 2002; Jiang and Zhang, 2020; Jia et al., 2015). While these dynamics may promote coordination or efficiency in certain contexts, they also contribute to corruption risks and institutional opacity (Pei, 2007; Wang and Zheng, 2020). Empirical research on grassroots officials in China remains limited, largely due to the opacity of personnel systems and the lack of disaggregated data. By compiling detailed career trajectories for over 1,600 city-level party secretaries, this paper provides rare grassroots evidence on the functioning of China's meritocratic apparatus. This paper reinforces the critical role that observable economic performance metrics play in determining promotion outcomes for Chinese officials. Leveraging the structural parallels between China's hierarchical administrative

[6] Some examples of inflated Chinese national or regional accounts have been reported: In 2017, the governor of Liaoning province publicly admitted to fabricating fiscal revenue and GDP figures by over 20% during the years 2011 to 2014. A report from *Xinhua Net*, a prominent state-run media outlet, quoted local officials who disclosed an "unspoken rule" within the system: "Local officials claimed that they could meet any targets set by higher authorities, regardless of difficulty, and that no target would remain unmet." In 2016, Tianjin's Binhai New Area revised its reported GDP from over 1 trillion yuan to 665.4 billion yuan, a nearly 30% reduction. This adjustment was attributed to revised statistical methods and the removal of inflated data. (*Sina News*) Shandong Province announced a revised GDP for 2018 at 6.6649 trillion yuan, a 12.8% decrease (982.1 billion yuan less) from the preliminary figure, following the fourth national economic census. The revision addressed issues such as duplicate reporting and inflated figures by adhering to stricter standards.

system and corporate governance—where higher-level authorities function analogously to a centralized board of directors—I model promotion dynamics using survival analysis, treating city officials as de facto heads of subsidiary units. Drawing on methods from labor economics, I further connect different cities through officials who rotate across jurisdictions, enabling the identification of individual-level contributions to target achievement while controlling for city-level unobservables.

Finally, this paper contributes to the literature on state-firm relations in China, a field with mixed findings on the extent of government influence over firms. Some studies argue that government interventions primarily target state-owned enterprises (SOEs), with minimal impact on private firms (Naughton, 2007; Lin and Milhaupt, 2013; Mueller et al., 2023). Others highlight substantial government influence over both SOEs and private firms through financial support, regulatory oversight, and political networks (Pearson, 2015; Huang, 2008; Piotroski et al., 2015). This influence also includes indirect mechanisms, such as fostering relationships between officials and business leaders, which shape decision-making across various enterprises (Nee, 1992; Jia et al., 2021; Feldman et al., 2021). The findings of this paper support the broader view, aligning with research that highlights significant government influence over the entire spectrum of Chinese firms. This paper also offers a possible perspective on the overproduction observed in some Chinese firms (Shen and Chen, 2017; Dong and Sun, 2022), by suggesting that such outcomes may be linked to politically driven resource allocation. In particular, the pursuit of short-term growth targets can encourage local officials to support firm-level output expansion even when market demand is weak, providing one potential mechanism through which sectoral overcapacity may arise.

Another key contribution of this paper is the construction of a novel, hand-collected dataset that links political, economic, and environmental information across multiple administrative levels in China. I manually compiled annual GDP growth targets and detailed career histories for over 1,600 city-level officials across 339 prefecture-level cities, using government work reports and official personnel announcements. These political data were then systematically merged with firm-level records from the China Industrial Enterprise Database and the China Industrial Enterprise Pollution Database, producing a unified panel that connects officials' performance incentives with firms' financial, energy, and environmental outcomes. This integrated dataset—assembled through extensive manual collection from multiple administrative and official sources—provides a unique empirical foundation for future research on political incentives, firm behavior, and environmental performance in China.

The structure of the paper is organized as follows: Section 2 provides an overview of the institutional background. Section 3 introduces the data. Sections 4 and 5 focus on officials and firms, respectively, detailing the empirical strategies and presenting the corresponding results. Section 6 explores output proxies, and Section 7 concludes.

2 Background

2.1 Target Management and Cadre Evaluation

The establishment of economic growth targets in China is deeply rooted in the country's historical and institutional evolution. Introduced in the 1950s as part of the Soviet-style planned economy, these targets initially functioned as mandatory elements of central planning. While the market-oriented reforms of the late 1970s dismantled much of the traditional planning framework, growth targets persisted as essential governance tools. Their continued relevance lies in their dual function: aligning local government objectives with national economic priorities and serving as performance benchmarks for local officials.[7]

The process of setting annual growth targets in China reflects the country's centralized administrative structure, following a hierarchical, top-down approach (Li et al., 2019). The central government sets national growth objectives, which are then cascaded to provincial and prefectural levels. Each tier of government formulates its own targets, often amplifying

[7]The link between economic outcomes and career advancement became particularly pronounced after Deng Xiaoping's 1992 Southern Tour, which ushered in an era of aggressive market reforms and rapid growth.

those of the higher level to demonstrate ambition and commitment.[8]

This target-setting process involves extensive negotiation and deliberation among officials, often taking months to finalize.[9][10] Although these targets are not legally binding, they carry significant weight as they encapsulate governmental priorities and provide a strategic framework for policy implementation. Figure 1 illustrates the trends in annual economic growth targets across different levels of government. It reveals a strong correlation between actual GDP growth and official targets, suggesting that target setting has served as an effective motivational tool. Moreover, lower-level governments have consistently set growth targets that are at least as ambitious as those of higher-level authorities, reflecting the top-down transmission of pressure embedded in the incentive system.

The cadre evaluation system institutionalizes the importance of growth targets by linking career advancement to quantifiable economic outcomes. Under the CCP's performance-based promotion framework, officials are assessed on a range of criteria, including political loyalty, ideological commitment, administrative capacity, and—crucially—economic achievements (Li and Zhou, 2005; Wang and Zheng, 2020). Regulatory documents such as the *Provisional Regulations on the Assessment of Party and Government Leading Cadres (1998)* codify this performance evaluation process. While qualitative assessments (e.g., leadership style or party discipline) remain important, quantitative indicators—particularly GDP growth—play a decisive role. Achieving or exceeding local growth targets enhances promotion prospects and

[8]For example, provincial growth targets frequently exceed national benchmarks by 10–30%, while prefectural targets typically surpass provincial ones by an additional 2–3 percentage points.

[9]In China, there is extensive bidirectional political communication between higher and lower levels of government (Chen et al., 2021). Before drafting local government work reports, higher-level governments hold economic work conferences to convey future policy intentions and engage in discussions about regional development within their jurisdictions. Following these meetings, governments at various levels begin drafting their respective work reports. During this process, workgroups from each government conduct in-depth research and gather relevant materials from subordinate organizations. Once an initial draft is prepared, feedback sessions are held where higher-level governments review the draft reports and provide feedback to ensure the reports are accurate and objective. Simultaneously, lower-level governments gain deeper insight into the policy directions of their superiors by analyzing the feedback and draft content. After multiple rounds of discussions, revisions, and consultations, the finalized government work report is submitted to the People's Congress for approval.

[10]A city's past economic performance plays a critical role in shaping its current growth target. This relationship is illustrated in Figure A1.

signals both administrative competence and political alignment with the Party's developmental agenda (Lyu et al., 2018). This performance-driven system has created strong incentives for local officials to pursue economic expansion, sometimes leading to excessive investment or even data manipulation. Yet, it has also contributed to rapid industrialization and infrastructure development.

2.2 State-Firm Relations

Growth targets ultimately matter insofar as they shape the behavior of firms—the primary engines of economic output. In China, the relationship between the government and enterprises is deeply embedded in the country's distinctive blend of state control and market-oriented reform. The government exerts substantial influence over businesses through regulatory frameworks, financial systems, and political connections, which are deeply ingrained across various sectors (Garnaut et al., 2018; Chen et al., 2019). For state-owned enterprises (SOEs), this influence is particularly direct and far-reaching (Jefferson and Rawski, 1994).[11] The state enforces alignment with its developmental goals through mechanisms like appointing party members to key managerial roles and embedding party committees within corporate governance structures (Liu and Zhang, 2018).

The government's influence extends beyond SOEs to private enterprises, which, while theoretically independent, remain subject to significant oversight and intervention. Access to critical resources such as financing, land use rights, and regulatory approvals often depends on maintaining favorable relationships with local or central authorities (Xiong, 2018; Piotroski and Zhang, 2014; Hsieh and Klenow, 2009).[12] These partnerships, whether voluntary or compelled, align private sector activities with national priorities. As a result, private firms operate within a system where their actions must partially conform to state

[11]SOEs are often viewed as instruments of national policy, tasked with achieving both economic and political objectives, such as maintaining employment, advancing strategic industries, and ensuring social stability.

[12]Moreover, private firms are frequently enlisted to support state-led initiatives, including the Belt and Road Initiative and industrial upgrading programs.

objectives (Huang, 2008; Song et al., 2011). This government-enterprise relationship plays a pivotal role in achieving economic goals. While SOEs are direct vehicles for state influence, private firms are also affected through a range of mechanisms, from production decisions to potential data manipulation (Chen et al., 2022; Martinez, 2022). Anecdotal evidence further supports the extent of the government's reach in shaping both enterprise behavior and economic outcomes.[13]

3 Data

This paper draws on three primary data sources: (1) official economic growth targets and realized GDP growth rates at various administrative levels; (2) appointment and promotion records of city-level political leaders; and (3) firm-level microdata on output, inputs and pollutants.

3.1 Growth Targets

China's growth targets are compiled from publicly available government documents and statistical bulletins. National targets are obtained from key policy documents such as the *Five-Year Plans for National Economic and Social Development* and the *Annual Report on the Work of the Government*, which outline macroeconomic goals including GDP growth, fiscal revenue, and social development indicators. Subnational targets—at the provincial

[13]The founder of Reading Motors, Li Guoxin, publicly accused Wang Xiao, the Party Secretary of Changle County, of pressuring the company to falsify industrial and sales output data worth 4.683 billion RMB since March 2022. This alleged data manipulation was likely driven by the need to meet local economic performance targets, a critical metric in China's cadre evaluation system (*Xinhua Net, 2023*). Similarly, in 2021, during the first round of statistical inspections, authorities held 278 individuals accountable for statistical violations, including 13 at the department level, 94 at the county level, and 162 at the township level. These inspections uncovered cases where local governments coerced enterprises into inflating their reported data by 20% to meet economic goals, leading to widespread data falsification (*Observer, 2021*). In 2018, the National Bureau of Statistics (NBS) publicly exposed five prominent cases of statistical violations, involving Tianjin's Binhai New Area, Inner Mongolia's Kailu County, Liaoning's Xifeng County, Shandong's Gaomi City, and Ningxia's Lingwu City. These cases highlighted severe falsification practices, such as fabricating reports, instructing enterprises to submit false data, and obstructing statistical law enforcement. Together, these incidents underscore the systemic pressures and challenges surrounding data reliability in the context of China's growth-oriented governance framework (*NBS, 2018*)

and prefectural levels—are sourced from regional five-year and annual development plans, which adapt national priorities to local contexts. These documents are officially published online. The dataset spans two decades and includes hand-collected data from 339 prefecture-level cities across all 32 provincial-level administrative regions. To measure actual economic performance, the stated targets are matched with realized GDP growth rates, primarily drawn from the China City Statistical Yearbook and local statistical reports.

3.2 City Leaders

A second data source consists of a unique panel of Chinese city party secretaries—hereafter referred to as governors—covering the same period and 339 cities for which growth target data were collected. The dataset includes detailed career and demographic information on over 1,600 distinct individuals. The data are sourced from official government websites and major state-run media platforms such as *Xinhua Net*, and include key demographic and professional attributes such as age, gender, birthplace, education, tenure length, and prior positions. A core feature of the dataset is the tracking of career trajectories, with a particular emphasis on promotion outcomes. Promotions are defined as upward transitions within the Chinese Communist Party's administrative rank system, following the criteria outlined in the *Regulations on the Management of Civil Servants' Positions and Ranks (2006)*. Each official's post-tenure outcome is classified into one of three categories: promoted, non-promoted (lateral transfer or sidelined), or retired.

3.3 Manufacturing Firms

The final source of data is at the firm level, encompassing two main components: (1) economic activity indicators, including sales, inventory, output, and input usage; and (2) environmental metrics, such as energy consumption and pollutant emissions. The economic data are primarily drawn from the China Industrial Enterprise Database (CIED), compiled from annual surveys conducted by the National Bureau of Statistics. It covers all state-owned enterprises and large non-state firms—defined as those with annual revenues above 5 million RMB before 2011 and 20 million RMB thereafter—across mining, manufacturing, and utilities, with manufacturing firms accounting for over 90% of the sample. The dataset spans from 1998 to 2014 and forms a large, unbalanced panel that includes detailed information on firm identity, ownership, employment, assets, sales, value-added, R&D expenditure, profits, and over 130 other variables.

The environmental, energy consumption and pollutant emissions data come from the China Industrial Enterprise Pollution Database (CIEPD). Developed by the Ministry of Environmental Protection, the CIEPD documents firms' energy usage and emissions for 27 types of pollutants, along with corresponding treatment measures. The CIEPD is matched with the CIED at the firm level, resulting in a panel of approximately 300,000 firm-year observations. This combined dataset includes key financial indicators alongside detailed metrics on energy use and pollutant emissions, spanning over 20 distinct industries within China's manufacturing sector. Additional details on data collection, cleaning and matching procedures, variable definitions, and examples are provided in Appendix A.

4 Political Incentives

This section documents the political incentives shaping the behavior of city governors in China. Meeting growth targets is strongly associated with improved promotion prospects, and the performance distribution of officials shows clear signs of strategic responses aligned with career advancement incentives.

4.1 A Survival Analysis

This subsection employs a survival analysis to examine the impact of economic performance on officials' promotion prospects.[14] Survival analysis is well-suited for this study for three

[14]The integration of survival analysis into political economy has also been explored extensively. (Acemoglu and Robinson, 2001) employed survival models to examine regime changes and transitions, highlighting how

primary reasons. First, economic performance indicators such as the GDP growth performance vary over time and influence promotion likelihood. Survival models can incorporate such time-dependent covariates (Therneau and Grambsch, 2000). Second, the study relies on panel data spanning approximately 20 years and covering multiple cities. This structure introduces censoring, as governors may leave the risk set without being promoted (e.g., due to retirement or remaining in office until the study ends). Survival analysis is specifically designed to address such incomplete observations (Klein and Moeschberger, 2006). Third, the panel structure of the dataset also results in delayed entry (Guo, 1993; Jenkins, 1995). Governors are appointed at different times throughout the observation period, meaning their "risk" of promotion begins at staggered points.[15] Formally, the observed time for each governor can be expressed as:

$$T_{\text{observed}} = \max(T_{\text{start}}, 0),$$

where T_{start} is the governor's appointment date relative to the study start time. Survival analysis accounts for delayed entry by adjusting the likelihood function to ensure that individuals are not considered at risk prior to their entry point.

The promotion data are recorded at discrete annual intervals, consistent with the structure of official appointments and personnel movements, which typically follow yearly cycles. This temporal structure justifies the use of discrete-time survival models and the discrete hazard function $h(j)$ represents the probability of promotion during the j-th interval, conditional on survival (non-promoted) up to its start. The survivor function $S(j)$ represents the probability that an official has not been promoted by time j. This study employs two discrete-

political institutions shape economic outcomes. (Svolik, 2012) used survival analysis to study the stability and collapse of authoritarian regimes. More directly related to this study, (Besley et al., 2008) analyzed how economic performance and institutional design influence the survival of autocratic regimes, showing that economic performance is a critical determinant of regime stability. Similarly, (Rauch and Evans, 2000) applied survival models to investigate how bureaucratic stability impacts policy implementation and effectiveness in developing nations.

[15]For example, a governor appointed in 2005 has a different observation window than one appointed in 2010.

time models—the logit model and the complementary log-log (cloglog) model—under both logarithmic and non-parametric specifications of the baseline hazard. To further account for unobserved heterogeneity, a frailty component is incorporated. Full model specifications and implementation details are provided in Appendix B. The logit specification models the promotion hazard as a function of both covariates and a baseline hazard term. Specifically, the log-odds of promotion at time t, conditional on not having been promoted earlier, is expressed as:

$$\log\left(\frac{h(t)}{1-h(t)}\right) = c(t) + \boldsymbol{\beta}'\mathbf{X}. \tag{1}$$

where $h(t)$ denotes the conditional probability of promotion in interval t, $c(t)$ captures the baseline hazard reflecting time dependence, and $\mathbf{X}$ includes key controls such as the GDP growth gap (actual minus target), governor age, prior experience in higher-level government positions, and educational attainment. The corresponding coefficient vector β reflects the relationships between these covariates and the promotion hazard.

Table 1, Column 1 reports estimates from a logit model predicting city governors' promotions based on key characteristics. Surpassing GDP growth targets significantly boosts promotion prospects—each one-unit increase in the actual-target gap raises the odds by 9.3%.[16] Age, prior experience, and education are also positively associated with promotion. Specifically, each additional year of age increases the odds by 3.1%; prior upper-level experience raises the odds by 15.1%; and higher education adds a 15.6% boost. Finally, longer tenures significantly increase promotion likelihood—each unit increase in log tenure raises the odds by 174%. This is evident, as in survival analysis, a longer "survival" duration typically indicates a higher likelihood of "failure" (promotion). Column 2 reports results from the complementary log-log model, which adopts a different functional form for the baseline hazard compared to the logit specification. Column 3 extends the model by incorporating a frailty term to account for unobserved individual heterogeneity. The frailty term is an

[16] $\exp(0.089) \approx 1.093$

individual-specific random effect that captures persistent unobserved characteristics making some officials systematically more or less likely to be promoted.

Figure 2 plots the likelihood of promotion as a function of officials' performance scores, measured by the gap between actual and target GDP growth. The results show a clear positive relationship: better performance is associated with a higher probability of promotion. Officials who fall short of their targets (gap < 0) consistently exhibit lower promotion rates—averaging below 5%—compared to over 10% for those who meet or exceed targets. Notably, there is a sharp discontinuity in promotion probability at the threshold where the performance score crosses zero. This jump suggests that officials who narrowly miss the target receive significantly fewer promotion opportunities than those who barely achieve it, highlighting strong incentives for marginal efforts to meet growth targets.

In summary, the survival analysis results suggest that officials with stronger economic performance are promoted more frequently and at a faster pace. [17] This partially explains their strong motivation to achieve growth targets. However, it is also possible that some officials benefit from favorable conditions by being assigned to already high-performing cities, allowing them to free-ride on existing growth momentum rather than actively contributing to performance improvements. Section 4.3 aims to explore this dynamic further.

4.2 Bunching Estimation

This section details the methodology employed to quantify the extent of politicians' deliberate efforts to meet economic growth targets. This analysis uses the bunching technique around points with discontinuities in incentives to elicit behavioral responses, a method initially introduced in the public economics literature by Saez (2010). The premise is that if politicians are driven by specific pressures to meet these targets, there will be a disproportionately high frequency of observations right at the threshold of achieving the target.

[17] Figures B1 and B2 analyze the relationship between tenure duration, promotion likelihood, and survival rates, differentiating between officials who meet growth targets and those who do not. The findings indicate that target-achieving officials experience faster career advancement.

To evaluate politicians' incentives to meet growth targets, the analysis focuses on the distribution of the actual-minus-target growth gap, as used in previous literature (Lyu et al., 2018; Chen et al., 2020). This gap is defined as the difference between the annual actual GDP growth and the target growth for the corresponding year. Let Z denote the support of the gap, and z^* the threshold where a politician precisely meets the target (i.e., gap $= 0$). To quantify the bunching mass B at z^*, the initial step involves estimating the counterfactual distribution $h_0(z^*)$ that would exist without the kink. This estimate is then compared to the observed distribution with the kink. The bunching mass B is defined as the difference between the observed spike in the distribution and the counterfactual density of the gap. Then normalize the bunching mass relative to the counterfactual in order to compare across various kinks with different counterfactual heights. Additionally, to account for potential diffuse bunching—where the concentration of observations is spread across an interval rather than centered at a single point—the bunching region is redefined to encompass a broader range. This approach accounts for situations where participants, such as politicians in this context, may not be able to precisely meet the target point z^*, resulting in a distribution of excess mass not just at z^* but within a vicinity around it.[18]

The estimation of the counterfactual without the kink follows the polynomial strategy proposed by (Chetty et al., 2011). This technique involves fitting a flexible polynomial to the observed distribution around the threshold of interest, while excluding the bins within the designated bunching region. By grouping individuals into bins of width δ, the resulting polynomial provides an estimate of $\hat{C}_j$. Formally, this is achieved by estimating the following model:

$$C_j = \sum_{i=0}^{p} \beta_i (z_j)^i + \sum_{i=z_L}^{z_U} \gamma_i \mathbb{1}\{z_j = i\} + \varepsilon_j. \quad (2)$$

[18] Specifically, define the bunching region as $z \in [z_L, z_U]$ surrounding a kink at z^*, where z_L and z_U represent the lower and upper bounds, respectively, of the "hump" area around z^*. This interval captures the spread of data points that are close to but not exactly at z^*, reflecting a more realistic scenario of target achievement that includes slight variations within a defined range.

C_j represents the observation count in bin j and p is the order of the polynomial used to fit these counts.[19] z_j is the gap mid point of bin j. The parameters z_L and z_U define the lower and upper bounds of the bunching region.[20]

The predicted counterfactual count in bin j in the absence of bunching is given by $\hat{C}_j$. The estimated excess bunching mass within the bunching region is then calculated as the difference between the observed and counterfactual bin counts:

$$\hat{B}_0 = \sum_{j:\, z_j \in [z_L, z_U]} \left(C_j - \hat{C}_j \right), \qquad \hat{C}_j = \sum_{i=0}^{p} \hat{\beta}_i (z_j)^i$$

$\hat{B}_0$ estimates the excess number of observations located at z^* due to the presence of a kink. $\hat{C}_j$ is the estimated count excluding the contribution of the dummies in the bunching region. To ensure comparability of results, normalize the bunching mass by the average counterfactual frequency in the excluded range:

$$\hat{b}_0 = \frac{\hat{B}_0}{\hat{C}_0}$$

Where $\hat{C}_0 = \left[\frac{z_U - z_L}{\delta}\right]^{-1} \sum_{j=z_L}^{z_U} \hat{\beta}_i(z_j)$. The standard error of the excess mass $\hat{b}_0$, is then determined through bootstrapping.[21] Panel A and Panel B of Figure C2 respectively illustrate the cases of sharp bunching (where $z_L = z_U = z^*$) and diffuse bunching (where $z^* \in [z_L, z_U]$) within the chosen excluded region. And the first lines of Panels A and B in Table 2 present the baseline estimates for the excess mass associated with sharp and diffuse bunching, respectively, along with their standard errors and t-statistics. These estimates of excess mass are statistically significant at the 1% level. An excess mass of 1.56 (2.03) indicates that there are 56% (103%) more observations at the target threshold than would be expected in

[19]This paper uses BIC criterion to determine the optimal degree of the polynomial (Bergolo et al., 2021)

[20]Following the Freedman–Diaconis rule (Freedman and Diaconis, 1981), the optimal binwidth δ which minimize the discrepancies between the height of histogram and real density is obtained by $2 \times IQR(x) * n^{-\frac{1}{3}}$. IQR stands for the interquartile range and n is the sample size.

[21]Specifically, generate 1,000 bootstrapped samples with replacement and calculate the standard deviation of the distribution of these estimates, which provides the standard error for $\hat{b}_0$.

the absence of the incentive. The analysis also accounts for potential biases arising from round-number bunching and naturally low density on the left side of the bunching region (Integration Constraint Correction). Detailed procedures are provided in Appendix C.

In conclusion, the distribution of the growth gap exhibits notable and unusual discontinuities at points where incentives shift, potentially indicating deliberate efforts by officials. The previous subsection establishes that meeting growth targets significantly improves promotion prospects. This finding translates into intentional efforts by officials to meet these targets, particularly when performance is close to the threshold.

4.3 Analysis Using Connected Samples

The earlier analysis does not account for the fact that the difficulty of achieving economic development targets varies significantly across cities and regions. Research also suggests that appointing officials to economically prosperous and strategically important cities often signals a pre-determined path for their career advancement, reflecting strong political connections or prior approval from higher authorities (Jiang and Zhang, 2020). Failing to account for these localized conditions when analyzing the economic impact of local leaders risks introducing bias (Jones and Olken, 2005).[22] These disparities underscore the importance of adopting an analytical framework that separates leader-specific contributions from city and time specific factors. A more refined analysis should focus on the extent to which a leader's efforts to close growth gaps influence their subsequent promotion outcomes.[23]

The periodic rotation of Chinese officials across cities enables this analysis, as the ma-

[22]Some cities possess inherent advantages—such as favorable geographic locations, historical industrial bases, or preferential economic policies—that naturally facilitate higher GDP growth. Leaders assigned to such cities may benefit from these advantages, and attributing economic success solely to their leadership would overstate their contribution. Conversely, leaders assigned to disadvantaged cities may appear less effective due to structural challenges beyond their control, rather than any deficiency in ability or effort. For example, consider an official who has already been informally designated for promotion and is subsequently reassigned by higher authorities to a city with strong economic momentum. In such cases, the official can meet growth targets with minimal effort by simply maintaining the existing trajectory of development.

[23]Moreover, the presence of laterally transferred officials could complicate the survival analysis, as these same individuals (movers) appear in the dataset multiple times after being reassigned to different cities. Treating each city-level tenure as an independent observation may obscure individual-specific traits and lead to bias in the estimated promotion probabilities.

jority of non-promoted leaders undergo lateral transfers, some serve as governors in other prefecture-level cities. These movements create a network that allows the separation of leader effects from city fixed effects.[24] In the sample, 15% of officials were "movers," meaning they served in multiple cities during their careers. These movers connect cities into a network, forming a "connected sample." In this network, a city is considered connected if at least one official who served there also held an equivalent position in another city. Based on this definition, 82.5% of the cities in the sample are classified as connected. A detailed breakdown of all connected groups is provided in Table D1 and Figure D1.

The methodology for estimating leader effects is adapted from the framework originally developed for analyzing linked employer-employee data. Abowd et al. (1999) (AKM) introduced a seminal two-way fixed-effects model that decomposes wage variation into worker-specific and firm-specific components. This approach exploits worker mobility across firms to identify these fixed effects, enabling the estimation of individual and firm-level contributions to wage outcomes.[25]

To evaluate leaders' contributions to closing the gap between actual GDP and target GDP, a three-way fixed-effects model is employed, drawing on the framework of Best et al. (2023) and Yao and Zhang (2015). The econometric specification is as follows:

$$\mathrm{Gap}_{ijt} = \mathbf{X}'_{ijt}\boldsymbol{\beta} + \theta_i + \psi_j + \gamma_t + \varepsilon_{ijt}. \tag{3}$$

In this specification, Gap_{ijt} denotes the GDP gap (actual GDP minus target GDP) for city j in year t during the tenure of leader i. X_{ijt} represents a vector of time-varying control variables, including the logarithm of total city population, provincial targets, and the annual average nightlight intensity for each city. θ_i captures the leader-specific fixed

[24] Specifically, comparing the same official's terms across different cities helps identify relative differences in city fixed effects, while comparing different officials' terms within the same city isolates relative differences in leader fixed effects.

[25] The AKM framework has been widely applied and extended across various domains, including managerial effects (Bertrand and Schoar, 2003) and leadership effects (Best et al., 2023; Yao and Zhang, 2015). These studies highlight the effectiveness of fixed-effects models in disentangling individual contributions from contextual effects, particularly in scenarios where mobility links individuals or entities across locations.

effect,[26] reflecting the relative contribution of leader i to closing the GDP gap. ψ_j accounts for city-specific fixed effects, capturing unobserved, time-invariant city characteristics such as geographic advantages or industrial composition. γ_t represents year-specific fixed effects, controlling for nationwide economic trends or macroeconomic shocks. The error term, ϵ_{ijt}, captures idiosyncratic variations.[27] The AKM variance decomposition provides insights into the relative contributions of four key components in explaining the variance of the growth gap: observed time-varying characteristics, leader-specific effects, city-specific effects, and the residual.

The variance decomposition results in Table 3 provide a detailed breakdown of the factors contributing to variation in the GDP gap. Leader fixed effects account for 28.1% of the variance in the growth gap, underscoring the significant influence of individual leaders in narrowing the gap. This substantial contribution reinforces findings from Jones and Olken (2005) and supports the earlier results of this paper, demonstrating the pivotal role leaders play in achieving economic growth targets. City fixed effects, by comparison, account for only 6.4% of the variance, suggesting that static, time-invariant city characteristics—such as geography or historical industrial bases—have a relatively minor influence. This indirectly strengthens the credibility of leaders' importance in meeting targets, as growth targets are largely artificial constructs, less directly tied to inherent city attributes. Observed time-

[26] Because leader tenures are typically short relative to the length of the city-level panel, estimates of leader fixed effects may be subject to considerable sampling variation. To mitigate this, an empirical Bayes shrinkage procedure is applied, adjusting each estimate toward the grand mean based on the number of observations per leader. (James and Stein, 1961)

[27] A key normalization is applied to address the inherent indeterminacy in the fixed-effects model (Abowd et al., 1999, 2002). This refers to the issue that fixed effects, such as leader and city effects, cannot be independently identified without imposing a normalization constraint due to their perfect collinearity with the model's constant term. Specifically, the sum of leader effects is constrained to zero:

$$\sum_i \theta_i = 0$$

This constraint ensures that leader effects are identified as relative contributions, avoiding collinearity with city or year effects. Without this normalization, leader effects would remain undefined, as any constant could be added to all θ_i and subtracted from ψ_j without altering the model's fit. Additionally, as previously mentioned, the analysis focuses on a connected sample of cities where leader mobility creates linkages across locations. This connectivity is essential for isolating the relative effects of leaders and cities.

varying characteristics, including time fixed effects and other control variables, explain approximately 29.4% of the variance. This result reflects the strong cyclical nature of growth targets, which are adjusted based on yearly economic conditions,[28] central government policy priorities, and the timing of leadership promotions.[29]

Table D2 and Figure D3 illustrate that leader-specific contributions to closing the growth gap are strongly and positively associated with promotion outcomes. Leaders ranking higher in terms of their contributions exhibit a significantly greater probability of promotion.

In conclusion, officials themselves play a significant role in determining whether growth targets are met, while the economic conditions of cities have a relatively weak relationship with target achievement. Even when officials are assigned to more economically developed cities, the findings suggest that they still need to actively strive to achieve their own contributions in order to enhance their promotion prospects. The results in this subsection are consistent with earlier findings, further reinforcing the conclusions of previous sections. Additionally, these findings mitigate concerns that earlier results might have been influenced by other confounding factors.

4.4 Summary

This section presented consistent evidence that growth targets significantly shape the career trajectories of local officials in China. First, survival analysis reveals a robust and statisti-

[28] Figure A1 illustrates how current-year target adjustments respond to last year's performance. While target revisions are generally positively associated with past performance, they tend to be conservative—particularly when the previous year's targets were missed, suggesting limited willingness to lower current targets further. Notably, there is no discontinuous shift in target setting at the threshold where the previous year's target was just met. This contrasts with the bunching pattern observed in Section 4.2 and reinforces the interpretation that the bunching arises from promotion incentives rather than cyclical or mechanical target-setting dynamics.

[29] For example, China's Five-Year Plans play a crucial role in shaping economic targets by setting overarching priorities and providing a framework for resource allocation. Growth targets, such as GDP growth, are often derived from the strategic goals outlined in these plans, ensuring alignment between long-term national objectives and short-term policy implementation. Additionally, the Five-Year Plans act as benchmarks for local governments, guiding regional economic policies and serving as a basis for performance evaluations. This alignment ensures that economic goals are not only ambitious but also strategically designed to address evolving challenges, such as technological innovation, environmental sustainability, and social welfare, while maintaining overall stability and growth.

cally significant relationship between target fulfillment and promotion: a one-unit increase in the actual-minus-target GDP gap raises the likelihood of promotion by approximately 9–10%, with high-performing officials also advancing more quickly. Second, bunching analysis detects clear discontinuities in the distribution of performance scores, with 56–103% excess mass at the threshold, suggesting strategic efforts by officials to meet or narrowly exceed growth targets. Third, the connected-sample analysis, based on a three-way fixed effects model, shows that leader-specific contributions explain over 25% of the variation in target fulfillment—substantially more than city fixed effects. Together, these results underscore that China's target-based evaluation system elicits strong behavioral responses from local leaders, who actively influence economic outcomes in pursuit of career advancement.

5 Impact on Firms

This section investigates how political incentives tied to GDP growth targets influence firm behavior. Using firm-level data, I examine whether pressure on local officials affects firms' operational decisions or reporting practices, focusing on changes in inventory and sales. In Section 6, I complement this analysis by using energy consumption and pollutant emissions as proxies for actual output.

5.1 Calculation of GDP in China

To understand how firm-level indicators might be influenced by officials' pressure to meet city-level GDP targets, it is essential to briefly explain how GDP in China's manufacturing sector is calculated. A distinctive feature of China's national accounting system is that GDP for the manufacturing sector—which contributed approximately 27% to 33% of total GDP in 2022—is calculated exclusively using the production approach.[30] Under this approach,

[30] According to the *China Gross Domestic Product Calculation Handbook (2001)* and the *China's System of National Accounts (2002)*, China employs a dual-method approach, combining the production approach (75% weight) and the income approach (25% weight). However, the income approach does not apply to the manufacturing sector, where GDP is calculated solely based on the production method.

GDP is based on the sum of value-added[31], which defined as the difference between total output and intermediate inputs[32]:

$$\text{Value-added} = \text{Total Output} - \text{Intermediate Inputs}.$$

In the manufacturing sector, total output is further decomposed into sales revenue, changes in inventory, and value-added taxes. Specifically, it is computed as:

$$\text{Total Output} = \text{Sales} + (\text{Ending Inventory} - \text{Beginning Inventory}) + \text{Value-added Taxes}.$$

Under the production approach to GDP calculation, overproduction—marked by excessive inventory accumulation—can inflate GDP figures. This raises the possibility that local governments may sometimes encourage such practices to meet growth targets. For instance, firms might be incentivized to overproduce, artificially increasing both inventory and total output.[33] Similarly, when firms boost sales, either through genuine market demand or by employing strategies like selling to customers with lower credit ratings, total output rises. Given these dynamics, this study focuses on two key firm-level indicators: sales and inventory changes.[34] From a financial reporting perspective, sales data is often subject to external auditing and verification, requiring documentation such as transaction records, invoices, and cash flow statements. This process enhances its transparency (Chen et al., 2020; Firth et al., 2011). In contrast, inventory adjustments are typically managed through internal accounting processes, making them less transparent and easier to manipulate for artificially inflating

[31] as reported by local enterprises

[32] Intermediate inputs refer to the costs of raw materials and services used in production, excluding expenditures on fixed assets and employee compensation.

[33] Although the manufacturing sector's GDP incorporates adjustments for fixed asset depreciation and inventory valuation to align with international accounting standards, these refinements primarily address discrepancies in inventory valuation and may not fully eliminate biases introduced when firms expand production to meet administrative targets (Xu and Liu, 2018).

[34] Early GDP calculations in China relied on indirect estimates derived from national income data under the Material Product System (MPS) inherited from the Soviet Union. Since the 1990s, direct calculations using firm-level and sector-level production data have been adopted to enhance accuracy and reliability. For instance, the value-added approach now incorporates detailed surveys of manufacturing outputs, intermediate inputs, and tax data collected directly from enterprises.

GDP figures (Gao et al., 2017).

5.2 Model Specification

To examine the impact of growth target pressures on firm outcomes, this paper employs a threshold-based strategy that leverages the discontinuity at the growth target threshold (Imbens and Lemieux, 2008). This approach assesses whether firms exhibit significant changes in behavior or reporting practices when the actual-target gap shifts from negative to non-negative, signaling that the economic growth target has been met. The model is specified as follows:

$$
\begin{aligned}
y_{ijkt} = \alpha + \beta D_{jt} + \gamma(1 - D_{jt}) \operatorname{Gap}_{jt} + \delta D_{jt} \operatorname{Gap}_{jt} \\
+ \mathbf{X}'_{it}\mathbf{\Gamma} + \phi_j t + \mu_i + \lambda_k + \varepsilon_{ijkt}.
\end{aligned}
\tag{4}
$$

In the model, y_{ijkt} denotes the outcome variable for firm i in city j industry k at time t, such as sales growth, production levels, or inventory changes. The binary indicator D_{jt} equals 1 if the actual GDP growth meets or exceeds the target ($\operatorname{Gap}_{jt} \geq 0$) for city j in year t, and 0 otherwise. The variable Gap_{jt}, representing the difference between actual and target GDP growth rates, serves as the forcing variable. Interaction terms are included to capture the slope of the forcing variable on either side of the threshold. Specifically, γ represents the marginal effect of the actual-target gap on the outcome variable when the target is not met ($D_{jt} = 0$), while δ reflects the marginal effect when the target is achieved ($D_{jt} = 1$). The model incorporates $\mathbf{X}_{it}$, a vector of firm-specific covariates (e.g., firm age, capital intensity, and ownership structure), to control for factors influencing firm behavior independently of growth targets. Additionally, μ_i accounts for firm fixed effects to control for time-invariant unobserved characteristics, while λ_k captures industry fixed effects to address sectoral heterogeneity. $\phi_r t$ represents city-specific time trends, capturing temporal variations unique to each city j. Instead of using year fixed effects—which control for uniform annual shocks across all cities but can absorb much of the variation in D_{jt}— this paper employs city-specific time trends.[35] The error term ε_{ijkt} represents unobserved heterogeneity and random shocks to the outcome variable. The primary parameter of interest, β, measures the discontinuous change in firm outcomes at the growth target threshold. This specification is designed to identify changes in firm behavior or reporting attributable to growth target pressures, although it does not necessarily imply causality.

5.3 Empirical Results

Panel A of Table 4 presents the empirical results using Inventory Changes as the dependent variable. For simplicity, I only retain the coefficient β as it is the sole variable of interest. The coefficients (β) across all five columns are positive, statistically significant ($p < 0.01$), and consistent in magnitude.

In Column (1), which excludes fixed effects or controls, the coefficient is 0.030, indicating that meeting the growth target is associated with a 3.0% increase in inventory changes on average. Introducing firm fixed effects in Column (2) raises the coefficient to 0.047 ($\approx$ 5%), suggesting that firm-level heterogeneity influences the observed effect. The magnitude stabilizes in Columns (3)-(5) as additional controls, including industry and city-specific time trends, are incorporated. City-specific time trends are used in Columns (4) and (5) instead of year fixed effects to avoid multicollinearity with β. The positive βs suggest a discontinuity or "jump" in firm-level indicators when the growth target is just met.

On the left side of Figure 4 corresponds to years when the city failed to achieve its annual growth target (gap < 0), while the right side reflects years when the target was met or exceeded (gap ≥ 0). The data demonstrate that firms in cities meeting their growth targets consistently exhibit higher inventory changes. A distinct jump is observed at the threshold (gap $= 0$), where inventory adjustments reach their highest levels. This discontinuity

[35] Since growth targets are often linked to national benchmarks or macroeconomic trends, year fixed effects can overlap significantly with D_{jt}, leading to multicollinearity and imprecise estimation of D_{jt}'s coefficient. In contrast, city-specific time trends preserve local variation while accounting for broader temporal dynamics within each city, enabling a more precise estimation of the effects tied to growth target implementation (Audretsch et al., 2015).

underscores the strong link between achieving growth targets and firm behavior regarding inventory management. To ensure the observed jump is not an artifact of splitting the graph into two segments, a complete and continuous plot of inventory changes against the actual-minus-target gap is provided in Figure E1. This complete curve confirms the presence of a clear and significant jump at gap = 0, reinforcing the robustness of the findings.[36], [37]

Panel B of Table 4 presents the results for ln(Sales). In Column (1), which excludes fixed effects and controls, the coefficient is negative. This likely reflects the overall decreasing relationship between sales and the actual-target gap, as the jump at gap = 0 is short-lived and not captured by a simple linear regression (see Figure 5). In Column (2), the inclusion of firm-level fixed effects results in a positive coefficient of 0.055 ($p < 0.01$), which remains consistent across subsequent specifications. Adding industry and city-specific time trends in later columns further refines the estimates. The positive and significant coefficients in Columns (2) through (5) indicate that meeting growth targets is associated with higher sales, once firm- and industry-specific effects are controlled for.[38]

In conclusion, although causality cannot be definitively established, firm-level variables closely linked to GDP calculations exhibit systematic shifts at the target threshold, potentially reflecting the pressures to meet growth targets. This pattern mirrors the bunching behavior observed among officials at the threshold, where promotion incentives intensify. The robustness of the firm-level results is confirmed across a range of tests, including sample restrictions, pseudo-thresholds, and alternative variable specifications. Further heterogene-

[36] Firms may adjust inventory levels, increase sales, or manipulate reporting to align with the economic goals set by local governments. Inventory accumulation, in particular, could serve as a strategy to artificially boost production indicators, contributing to the observed increase in GDP. This aligns with the broader context of political incentives in China, where local officials may intervene in firm operations to meet growth targets. However, the evidence does not establish a causal relationship between political incentives and firm production.

[37] The relatively low R^2 values are typical for RDD-type regressions, as the focus is on identifying local effects at the threshold rather than explaining overall variance.

[38] Standardizing by firm size is essential in this analysis. In economically advanced regions of China—where private sector activity tends to be more dynamic and competitive—firms are generally smaller on average. Given that the unit of observation is the individual firm, failing to account for size would disproportionately weight larger firms, which naturally exhibit higher absolute values across key variables. Normalization ensures that firm-level outcomes are comparable across heterogeneous economic environments and mitigates potential biases arising from regional differences in firm structure.

ity analysis reveals that these observed jumps are more pronounced in regions facing higher growth pressure and in years where cities are closer to meeting their targets. Detailed descriptions of robustness checks and heterogeneity analyses are provided in Appendix E and Appendix F.

6 Mechanism

The previous section provided clear evidence that meeting growth targets is strongly positively correlated with firm-level GDP-related indicators, such as changes in inventory. As expected, this relationship is even more pronounced among groups with higher incentives. However, a critical question remains: To what extent do the observed deviations in output-related indicators at the threshold reflect genuine economic activity, and how much is attributable to data manipulation?[39] Rather than questioning the authenticity of the data itself, this paper focuses on determining how much of the observed growth pressure translates into actual economic activity at the firm level. To address this, the study adopts an approach based on the relationship between firms' inputs and outputs. The central assumption is that while firms may have incentives to manipulate output-related indicators, they are less likely—or find it much harder—to manipulate observable metrics that are closely tied to output but difficult to falsify or offer little incentive for manipulation (Zeng and Zhou, 2024). Examples include energy consumption and pollution records: the former represents inputs that are less subject to manipulation,[40] while the latter reflects "undesirable outputs" that

[39] China's data—whether official national accounts or firm-level financial reports—has long been scrutinized by both external observers and academics. Prior researches have examined data reliability from various perspectives, often suggesting adjustments to enhance credibility (Chen et al., 2019; Firth et al., 2011).

[40] In China, labor quantity is not a reliable variable input for measuring firms' economic activities. This is largely due to the widespread use of "labor dispatching," where companies hire a significant portion of their workforce through third-party agencies. These dispatched workers are formally employed by labor dispatch firms but are assigned to work at industrial enterprises. Although they are technically considered temporary employees, many remain in the same firms for extended periods. Companies rely on dispatched workers to enhance workforce flexibility and reduce labor costs. However, because these workers are not officially registered as employees of the firms they work for, they are not included in employment surveys conducted by the National Bureau of Statistics (NBS). As a result, firms may systematically underreport their actual workforce size. (Brandt et al., 2014)

occur alongside production. These inputs or "undesirable outputs" are typically unrelated to growth targets and are collected by different agencies. The first subsection delves into the details of pollution and input-related data, while the second subsection examines the extent to which concerns about data manipulation are justified.

6.1 Pollution and Energy Consumption

Pollution and energy consumption are closely tied to the production activities (Shapiro and Walker, 2018), provoking the question: Are energy usage and pollutant generations also influenced by whether growth targets are met? Moreover, can a similar discontinuity at the target threshold be observed for these indicators? The Chinese Ministry of Ecology and Environment conducts annual surveys of manufacturing firms, collecting data on various water and air pollutants, energy usage, and certain raw material inputs. Notably, many of these firms overlap with those in the CIED dataset. Given the varying combinations of inputs used and pollutants generated across firms, it is essential to standardize or aggregate energy consumption and pollutants to enable meaningful inter-firm comparisons and mitigate issues of intercorrelation (Bu and Shi, 2021; Rijal and Khanna, 2020).[41] A detailed discussion of these distinctions and the standardization methods is provided in Section 3 and in Appendix A.

Panel A reports energy consumption in logarithmic form, covering industrial water usage, coal, natural gas, and diesel consumption. Panel B shows pollutants generated, including wastewater, sulfur dioxide, smoke and dust, and ammonia nitrogen. The final columns of both panels aggregate total energy consumption and pollutant generations for each firm. The results reveal a strong positive correlation between energy consumption, pollutants, and whether the city meets its economic targets. A similar jump at the target threshold is observed, mirroring the pattern seen in firm-level inventory changes and outputs. This suggests that, even if data manipulation occurs, at least part of the observed jumps in

[41] For pollutants, distinguishing between those generated and those discharged is particularly important, as firms employ different pollution treatment technologies that can significantly reduce emissions.

inventory changes and firm output at the threshold reflect real economic activity, rather than being solely driven by manipulation.[42] Figures G1 and G2 provide graphical representations of the results reported in Table 5. All energy consumption and pollutant emission variables exhibit a clear jump at the threshold and follow a similar overall pattern across indicators.

Table 5 presents firm-level results for energy consumption and pollutants, employing the same framework as in Section 5.

6.2 Disentangling "True" Output from Data Manipulation

This section employs pollution and energy consumption data as proxies for output, much like prior studies that have used nighttime light data as a proxy for GDP (Zeng and Zhou, 2024; Henderson et al., 2012; Martinez, 2022). The objective is to quantify how much of the observed jump in output at the target threshold is likely attributable to genuine economic activity versus data manipulation. Suppose the total output Y_{it} for firm i at time t can be decomposed into two components:

$$Y_{it} = Y_{T,it} + Y_{M,it} = f_{it}(L_{it}, K_{it}, e_{it}) + g_{it}(D_{jt}). \tag{5}$$

where $Y_{T,it}$ represents true output within a standard production function framework, driven by genuine economic activity and inputs such as energy (e), labor (L), and capital (K), without manipulation. In contrast, $Y_{M,it}$ represents manipulated output, reflecting the artificially inflated portion of Y_{it} arising from behaviors motivated by the target D_{jt}. Let e_{it} denote energy usage and p_{it} denote pollutants generated, which can also be interpreted as undesirable outputs for firm i in year t (Atkinson and Dorfman, 2005). The variable D_{jt}, as defined earlier, is a target threshold dummy that equals 1 if the city j meets its growth target in year t, and 0 otherwise.

[42] Notably, the coefficients for energy consumption and pollutants remain relatively stable and fall within a narrow range. This indicates that the relationship between most inputs, undesirable outputs, and overall output is constrained, which is an important consideration for interpreting the results in the subsequent subsection.

The key assumption is that the relationship between energy usage and pollution, $\frac{\partial p_{it}}{\partial e_{it}}$, is proportional to the relationship between energy usage and true output, $\frac{\partial Y_{T,it}}{\partial e_{it}}$, with a proportionality constant $\alpha > 0$:

$$\frac{\partial p_{it}}{\partial e_{it}} = \alpha \cdot \frac{\partial Y_{T,it}}{\partial e_{it}}$$

This implies that energy influences both true output and pollution at a constant rate. Additionally, manipulated output ($Y_{M,it}$) depends solely on the target threshold (D_{jt}) and is independent of energy usage:

$$\frac{\partial Y_{M,it}}{\partial e_{it}} = 0$$

This reflects the assumption that officials only have incentives to manipulate output for meeting growth targets. Under these assumptions, it is straightforward to estimate the proportion of the observed output "jump" at the target threshold that can be accounted for by corresponding increases in firm-level energy consumption or pollutant emissions. (Details are provided in Appendix G)

Table 6 presents the estimates of "true" output proportion ($\gamma_1 \sim \gamma_4$),[43] representing the proportion of true output by industry type. Here, γ_1 uses a PCA-derived latent measure of true output; γ_2 relies primarily on Chinese pollution data while imposing a constant proportionality parameter α between pollution and true output; γ_3 calibrates the output–energy relationship using U.S. industrial data; and γ_4 further uses U.S. industry-level pollution data to allow α to vary across industries. Since the value of γ is sensitive to the scale of the vari-

[43] To estimate γ_1, reliable true output data is required. In the absence of such data, this paper employs Principal Component Analysis (PCA) to estimate a latent output variable. Using data on energy consumption, labor, and capital, PCA captures the common variance among these input and observable variables, treating this latent variable as the true output regardless of potential manipulation (Kao et al., 2011). This approach effectively identifies the underlying factors that collectively represent unobserved true output. The output elasticity of energy consumption is then estimated using a Levinsohn-Petrin (Levinsohn and Petrin, 2003) production function approach, with the latent output serving as the dependent variable. To estimate the components necessary for calculating γ_2, regression-based techniques are employed to approximate the partial derivatives that define the relationships between output, energy consumption, pollution, and target completion. These estimations are conducted using two-way fixed effects models. γ_3 leverages the observed relationship between energy consumption and industrial output in the U.S. to infer the true output proportion of China. For γ_4 U.S. industrial pollution data are employed to generate a more accurate approximation of α (Levinson, 2009; Shapiro and Walker, 2018). This approach allows for variation in the elasticity between output and pollution across industries.

ables, all variables are standardized to have a mean of zero and a standard deviation of one to ensure comparability and consistency in the analysis. Only industries where output exhibits a significant jump at the target threshold are included in this table. The final row provides results for the full sample, which includes industries without significant jumps. Overall, approximately 10–15% of the output jump at the threshold may be attributed to potential manipulation. [44] Put differently, around 10–15% of the observed firm-level output jump at the threshold cannot be explained by corresponding changes in energy consumption or pollutants generated. This conclusion holds if the energy and pollution data are reliable.[45]

This aligns with findings from previous related literature.[46] For nearly every industry, γ_2 values are higher than γ_1, and both share similar patterns. γ_3 and γ_4 are supplements to γ_1 and γ_2. γ_3 mirrors the definition of γ_1, but uses U.S. industrial output—which is assumed to be free from manipulation—as the benchmark. The relationship between energy consumption and output in the U.S. is applied in place of the corresponding Chinese data to estimate the proportion of real output. The results show a strong similarity between the two contexts. γ_4 further relaxes the assumption on α, allowing it to vary by industry to account for sector-specific output–pollution elasticities, while taking the normalization such that the overall sample's α remains equal to 1. Missing values indicate industries where severe data gaps in energy or pollution data prevent the calculation of these estimates.

Industries where γ_2 significantly exceeds γ_1, such as the *Textile Industry* (70.2% vs. 88.2%) and *Non-Ferrous Metal Smelting* (47.7% vs. 87.9%), suggest that pollution is a

[44] While it is plausible that 10–15% of the observed discrepancy falls within a reasonable margin of error and may not constitute manipulation per se, the primary objective of this analysis is not to quantify the extent of manipulation. Rather, the aim is to assess the extent to which political incentives translate into real economic effects.

[45] However, Ghanem and Zhang (2014) highlighted that even pollution data in China is subject to potential manipulation.

[46] Li et al. (2024) estimated that 5–10% of China's GDP data may be overstated due to fabrication, while Firth et al. (2011) observed significant discrepancies between the aggregated local GDP figures reported by certain Chinese cities and regions and the national GDP data, with local reports potentially overstating actual values by 15–20%. Similarly, Chen et al. (2019) highlighted that since the mid-2000s, the National Bureau of Statistics has consistently revised down local governments' reported GDP figures by an average of 5%. Holz (2014) noted that since 1997, the aggregated provincial GDP in China has often exceeded the national GDP, with the gap reaching as high as 19.3% by 2004.

stronger indicator of true output than energy usage, likely due to inefficient technologies and reliance on pollutant-heavy energy sources like coal. In contrast, industries where γ_1 closely aligns with γ_2, such as *Coal Mining and Washing* (56.1% vs. 56.9%) and *Non-Metallic Mineral Products* (93.6% vs. 94.9%), reflect a direct relationship between energy use and pollution, where energy consumption translates more directly into pollutant generation due to the nature of production processes. Certain industries, such as *Leather and Related Products* (41.2% vs. 49.5%) and *General Equipment Manufacturing* (33.4% vs. 26.5%), exhibit relatively low true output proportions ($< 60\%$), indicating stronger incentives for manipulation or weaker alignment of energy and pollution proxies with actual economic activity. For example, *Leather and Related Products* may rely on low energy and pollution intensity, making these proxies less reflective of output, while *General Equipment Manufacturing* often employs advanced, energy-efficient technologies that reduce the correlation between these proxies and true economic activity. Overall, while low true output proportions in some industries may suggest significant data manipulation or weak proxy alignment, the broader findings highlight that growth targets often lead to substantial real economic activity. Firms appear to have meaningfully expanded production, whether driven by political pressures or favorable economic conditions.

7 Conclusion

This study examines how local officials' incentives influence economic outcomes within China's performance-driven governance system. The findings highlight the substantial impact of growth targets on both political and economic behaviors. Using a combination of bunching analysis and survival models, the study reveals that local officials strategically adjust economic performance metrics to meet growth targets, particularly when promotion incentives are strong. Achieving targets provides significant advantages for officials' career advancement. At the firm level, the evidence shows a clear relationship between political and growth pressures and GDP-related indicators. These findings are robust across alternative dependent variables and specific subsamples. When considering quarterly GDP as an midpoint performance measure, officials who fall behind early in the year tend to intensify efforts in the remaining months to catch up, further strengthening the link between meeting targets and firm-level economic activity. Moreover, using firm-level energy consumption and pollution data as reference points, the study finds that a significant portion of the output jump at the target threshold can be explained by corresponding changes in pollution and energy usage. This suggests that local officials' actions go beyond simple data manipulation and have tangible effects on real economic activity.

The findings of this study hold significant policy implications. As Rong (2013) illustrates, the core operational logic of a hierarchical, pressure-driven governance system involves the top-down transmission of directives through the exertion of pressure, coupled with the superior's absolute authority to reward or punish subordinates. This structure inevitably translates the superior's often "vague" intentions into specific and narrowly focused targets.[47] Without such quantifiable measures, it becomes challenging for superiors to assess subordinates' performance or ensure the effective transmission of their intentions. However, in the absence of robust oversight and accountability mechanisms, such systems are prone to unintended consequences and opportunities for rent-seeking. This paper provides valuable empirical insight to deepen our understanding of these dynamics.

This study has several limitations, highlighting opportunities for future research. Both the bunching analysis and threshold based specification, while valuable, lack the ability to establish definitive causal relationships. While we observe firms responding to growth pressures, the specific mechanisms through which governments influence firms remain unclear. Future research could explore this in greater depth, leveraging more direct data linking government actions to firm behavior.

[47]For instance, directives like "developing the economy" are often translated into GDP growth targets, while "managing the COVID-19 pandemic" is reduced to metrics such as infection rates. Similarly, "emissions reduction" is measured by CO_2 emissions, and "poverty alleviation" is quantified by the number of households removed from poverty.

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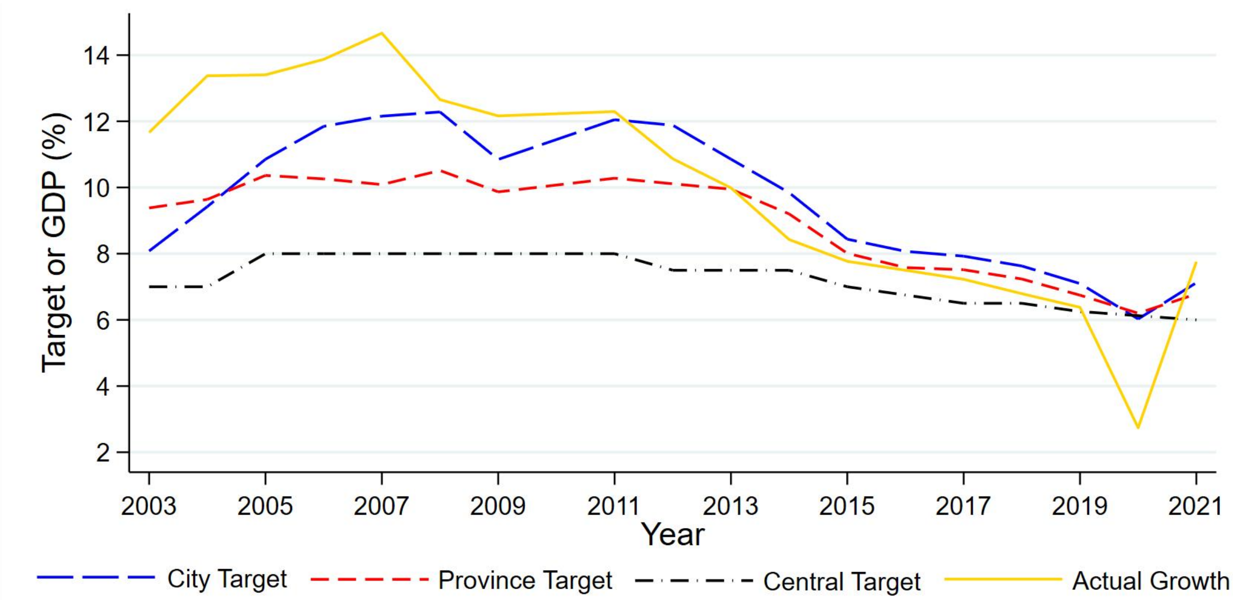
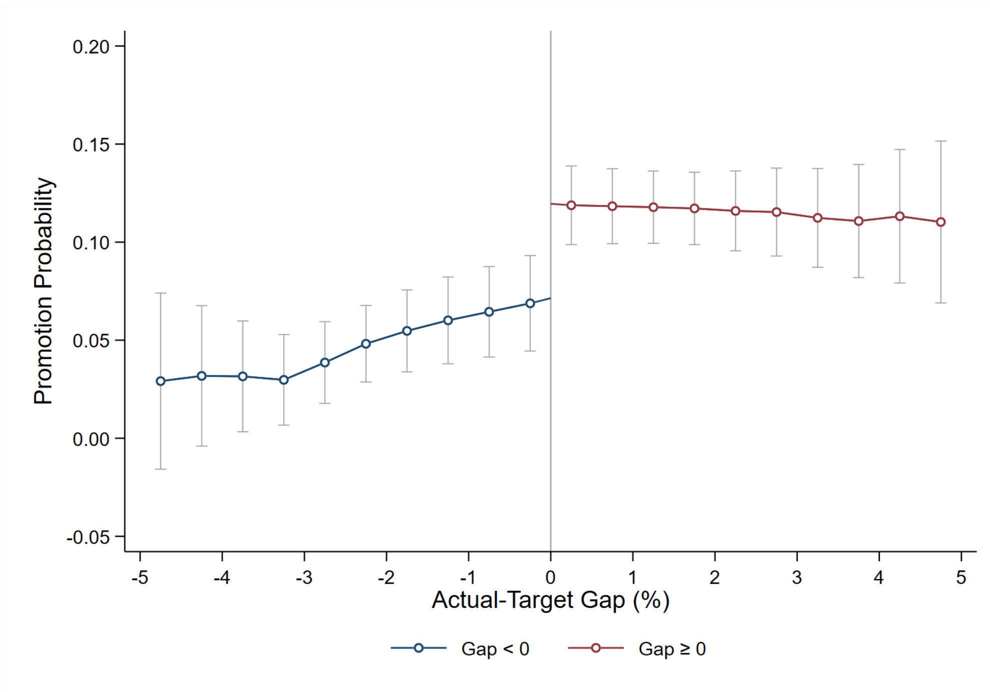
Figure 1: Time Trend of Economic Growth Targets And Overweights



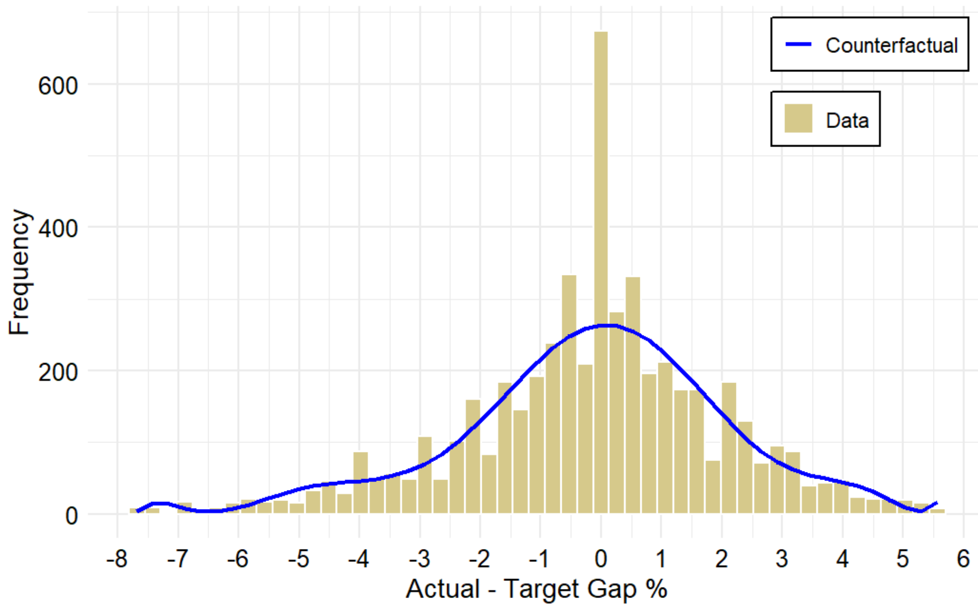
Notes: This figure presents the weighted averages of city-level and provincial annual growth targets over the past 20 years, along with the national growth target and the actual (constant prices) GDP growth rate. The weights are based on each city's or province's respective GDP share. The figure reveals a strong alignment between growth targets and actual performance, particularly in earlier years when actual growth frequently exceeded targets. Since 2013, however, both targets and actual growth have become less ambitious, consistent with findings in the literature suggesting a reduced emphasis on economic performance in official evaluations after this period (Holz, 2014; Chen et al., 2018). The figure also clearly demonstrates the "top-down amplification effect" discussed earlier.

Figure 2: Promotion Probability by Actual-Target Gap



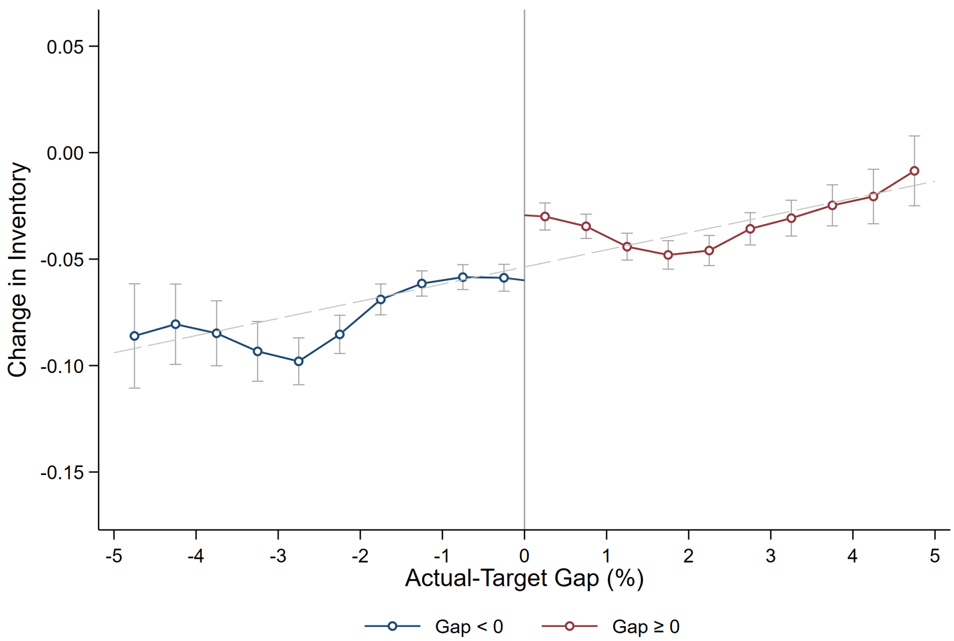
Notes: Figure 2 illustrates the (unconditional) relationship between officials' promotion probabilities and their economic performance, measured by the actual-target gap. The figure reveals that promotion probabilities increase as the actual-target gap widens, with officials who exceed their targets consistently demonstrating higher promotion probabilities than those who fall short. A noticeable jump in promotion probabilities occurs at Gap = 0, marking the transition from underperformance to meeting targets.

Figure 3: Distribution of the Actual–Target Gap and Counterfactuals



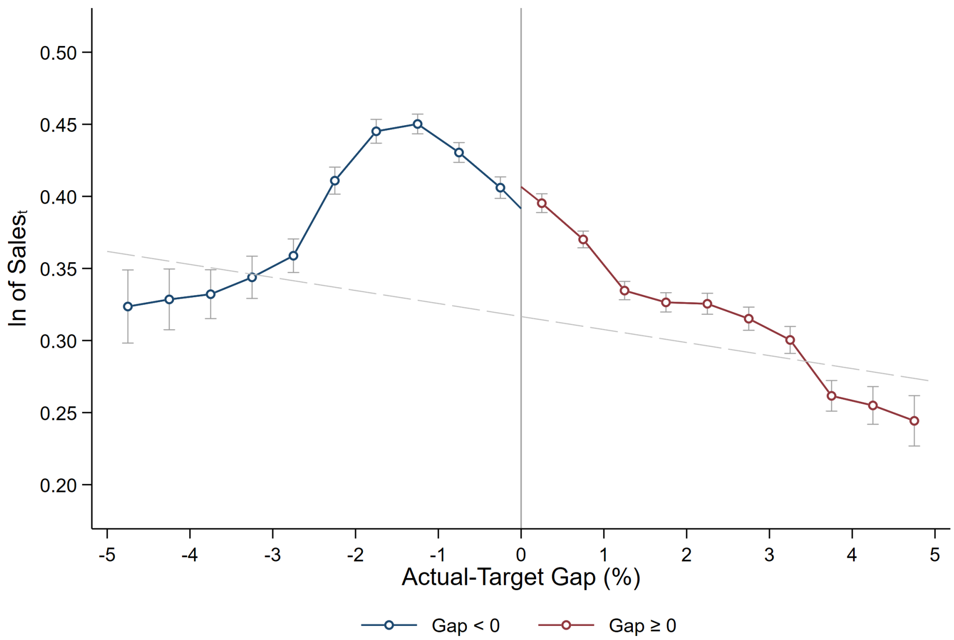
Notes: Figure 3 presents the empirical distribution of the actual-minus-target GDP growth gap (histogram), overlaid with the estimated counterfactual distribution (blue line). The corresponding excess mass estimates and standard errors are reported in Table 2. Figure C1 presents the counterfactual distribution estimated with additional controls, including indicators for round-number gaps and localized density spikes ("nearby kinks"), to account for potential confounding patterns in the observed distribution.

Figure 4: Change in Inventory by Gap



Notes: This figure plots the firm-level ln(change in inventory) as a function of the city's economic performance (actual-target gap). The line one the left represents years when the city failed to meet its annual target, while the right side represents years when the city at least achieved its target. The figure shows that cities meeting their targets not only exhibit higher levels of inventory changes overall but also display a clear jump at the threshold (gap = 0). At this point, the change in inventory reaches its peak. Because separating the graph into two sides can artificially introduce or amplify a discontinuity, I have also plotted a complete curve for change in inventory in Figure E1.

Figure 5: Ln of Sales by Gap



Notes: This figure plots the firm-level ln(sales value) in year t as a function of the actual-target gap. Overall, sales exhibit a decreasing trend as the gap widens. Similarly to Figure 5, the red line on the left represents years when the city failed to meet its annual target, while the right side corresponds to years when the city at least met its target. A jump at gap $= 0$ is still visible, although it quickly dissipates. Same as above, in Figure E2, I have plotted the complete curve without imposing a discontinuity at gap $= 0$.

Table 1: Survival Analysis with Logarithmic Baseline Hazard

	Promotion Dummy		
	(1)	(2)	(3)
	Logit	Complementary log-log	Generalized Gamma
Actual-Target Gap	0.089***	0.080***	0.098***
	(0.020)	(0.018)	(0.021)
Age of Governor	0.031***	0.028***	0.045***
	(0.012)	(0.010)	(0.014)
Prior Experience	0.141***	0.125***	0.157***
	(0.046)	(0.042)	(0.054)
Education Level	0.145**	0.126**	0.106
	(0.068)	(0.061)	(0.079)
ln(spell length)	1.008***	0.917***	1.378***
	(0.080)	(0.072)	(0.135)
Observations	4507	4507	4504

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: This table presents survival analysis results using different models. The dependent variable is a promotion dummy indicating whether a politician is promoted. This result is based on a parametric baseline hazard (logarithmic form). As expected, the natural logarithm of tenure length positively correlates with promotion likelihood, indicating that longer tenures are beneficial for officials' career advancement. The first column presents the results of the logit model, the second column corresponds to the cloglog model, and the third column shows the results from the cloglog model incorporating individual-level variability. This individual heterogeneity is assumed to follow a gamma distribution. Next, regarding the covariates: Actual-Target Gap is defined as the city's annual actual growth rate minus its target growth rate. Note that this measure is simply the difference between the two values and is not represented as a dummy variable, such as when the difference is greater than or equal to zero. The age of the governor is also included as a covariate. Prior Experience indicates whether the official has previously worked in higher-level government positions. For instance, an official transferred from a provincial government department to a prefecture-level city government would be classified as having prior experience. Conversely, officials classified as "grassroots" have spent their careers working locally, starting from lower-level positions. Education level reflects the governor's educational attainment. As mentioned in the data section, officials may have three distinct types of educational backgrounds, and the variable used here is the sum of these three categories.

Table 2: Estimates of Excess Mass

Panel A: Sharp Bunching			
	Excess Mass $\hat{b}$	Standard Error	t-statistic
Sharp	1.56	0.164	9.51***
Sharp: round #	1.518	0.287	5.29***
Sharp: correction	1.381	0.314	4.40***

Panel B: Diffuse Bunching			
	Excess Mass $\hat{b}$	Standard Error	t-statistic
Diffuse	2.033	0.378	5.38***
Diffuse: round #	1.999	0.375	5.33***
Diffuse: correction	1.543	0.774	1.99**

Notes: Sharp bunching sets the threshold exactly where the gap equals zero, whereas diffuse bunching defines an interval as the bunching region. For this study, the interval $[-0.2\%, 0.5\%]$ was selected based on visual inspection of the distribution. "Round number" controls for whether the gap is a multiple of 0.5%. "Correction" refers to the integration constraint correction detailed in Appendix C. Figure C2 provides a graphical representation of the estimated excess mass

Table 3: Variance Decomposition of the GDP Gap

Component	**Proportion of Total Variance**
Leader fixed effects	0.281
City fixed effects	0.064
Observed time-varying characteristics	0.294
Residual	0.362

Table 4: Impact on Firms

Panel A: Inventory Changes

	ln of Inventory$_t$ − ln of Inventory$_{t-1}$				
	(1)	(2)	(3)	(4)	(5)
GDP Meets/Exceeds Target	0.030***	0.047***	0.063***	0.062***	0.061***
	(0.005)	(0.007)	(0.008)	(0.008)	(0.008)
Firm FE		✓	✓	✓	✓
Industry FE			✓	✓	✓
City Trends				✓	✓
Controls					✓
Observations	214769	214769	191275	191275	189981
R^2	0.0005	0.0008	0.0014	0.0014	0.0017

Panel B: Sales

	ln of Sales$_t$				
	(1)	(2)	(3)	(4)	(5)
GDP Meets/Exceeds Target	-0.037***	0.055***	0.044***	0.044***	0.044***
	(0.005)	(0.004)	(0.004)	(0.004)	(0.004)
Firm FE		✓	✓	✓	✓
Industry FE			✓	✓	✓
City Trends				✓	✓
Controls					✓
Observations	309535	309535	281408	281408	279518
R^2	0.0017	0.0029	0.0036	0.0037	0.0098

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: This table presents the empirical results for two firm-level indicators. "GDP Meets/Exceeds Target" is an indicator equal to one when a city's actual GDP growth rate meets or exceeds its announced growth target, and zero otherwise. Panel A uses change in inventory as the dependent variable, while Panel B uses ln(sales). As discussed in Section 5.1, change in inventory is defined as the difference between (ln of) end-of-period and (ln of) beginning-of-period inventory. Both dependent variables are standardized by firm size, defined as the ln(total assets) in year $t-1$. The results in both panels are shown across five columns. Column 1 includes no fixed effects or controls, Column 2 adds firm fixed effects, Column 3 incorporates industry fixed effects, Column 4 includes city-specific time fixed effects, and Column 5 adds firm-level controls. The results across Columns 2 to 5 are notably consistent, with little variation compared to Column 1. Firm-level controls include variables such as firm age, employment and ownership type (e.g., private, state-owned, or foreign).

Table 5: Pollution and Energy Consumption

Panel A: Energy Consumption

ln of	*Industrial Water Usage*	*Coal Consumption*	*Natural Gas*	*Diesel Consumption*	*Total Energy*
	(1)	(2)	(3)	(4)	(5)
GDP Meets /Exceeds Target	0.081*** (0.004)	0.038*** (0.002)	0.026*** (0.003)	0.031*** (0.004)	0.071*** (0.004)
Firm FE	✓	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓	✓
City Trends	✓	✓	✓	✓	✓
Observations	342158	276915	276915	276915	276915

Panel B: Pollutants

ln of	*Wastewater*	*Sulfur Dioxide*	*Smoke and Dust*	*Ammonia Nitrogen*	*Total Pollution*
	(1)	(2)	(3)	(4)	(5)
GDP Meets /Exceeds Target	0.031*** (0.003)	0.024*** (0.002)	0.043*** (0.004)	0.015*** (0.004)	0.098*** (0.007)
Firm FE	✓	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓	✓
City Trends	✓	✓	✓	✓	✓
Observations	304899	276915	276915	195904	193854

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: This table presents the firm-level results for energy consumption and pollutants. Panel A examines the impact of target achievement on energy consumption, including industrial water usage, coal, natural gas, and diesel consumption. Panel B reports the results for pollutant emissions, including wastewater, sulfur dioxide (SO_2), smoke and dust, and ammonia nitrogen. The final columns in both panels show the total energy consumption and total pollutants generated by each firm, obtained by aggregating all types of energy consumed and pollutants emitted. The results indicate a significant positive correlation between energy use, pollutant emissions, and target achievement.

Table 6: True Output Proportion by Industry Type

Industry Name	**Obs.**	γ_1 (%)	γ_2 (%)	γ_3 (%)	γ_4 (%)
Agricultural and By-Product Processing	39,665	71.2%	84.5%	–	–
Food Manufacturing	21,394	82.9%	95.8%	98.1%	98.9%
Beverage Manufacturing	17,993	–	87.3%	–	90.1%
Textile Industry	51,811	70.2%	88.2%	86.6%	88.9%
Leather and Related Products	8,723	41.2%	49.5%	58.9%	49.9%
Coal Mining and Washing	19,439	56.1%	56.9%	–	–
Wood Processing and Bamboo Products	8,459	44.3%	–	43.6%	–
Paper and Paper Products	29,027	79.4%	96.8%	83.0%	97.4%
Chemical Materials Manufacturing	72,249	88.7%	96.1%	71.0%	96.9%
Rubber Products	7,398	65.3%	–	99.9%	–
Non-Metallic Mineral Products	80,311	93.6%	94.9%	66.2%	94.6%
Ferrous Metal Smelting	21,321	56.9%	72.4%	48.0%	72.3%
Non-Ferrous Metal Smelting	15,985	47.7%	87.9%	85.2%	87.8%
Metal Products	22,305	33.6%	89.2%	24.6%	90.8%
General Equipment Manufacturing	24,815	33.4%	26.5%	35.2%	25.2%
Electrical Machinery Manufacturing	13,046	42.2%	45.0%	59.9%	44.0%
All Sample		90.1%	84.9%	84.6%	84.9%

Notes: This table presents the two metrics of true output proportion, γ_1 and γ_2, across different industries, along with the γ_3 and γ_4 based on U.S. industrial data. All values are expressed in percentage terms.